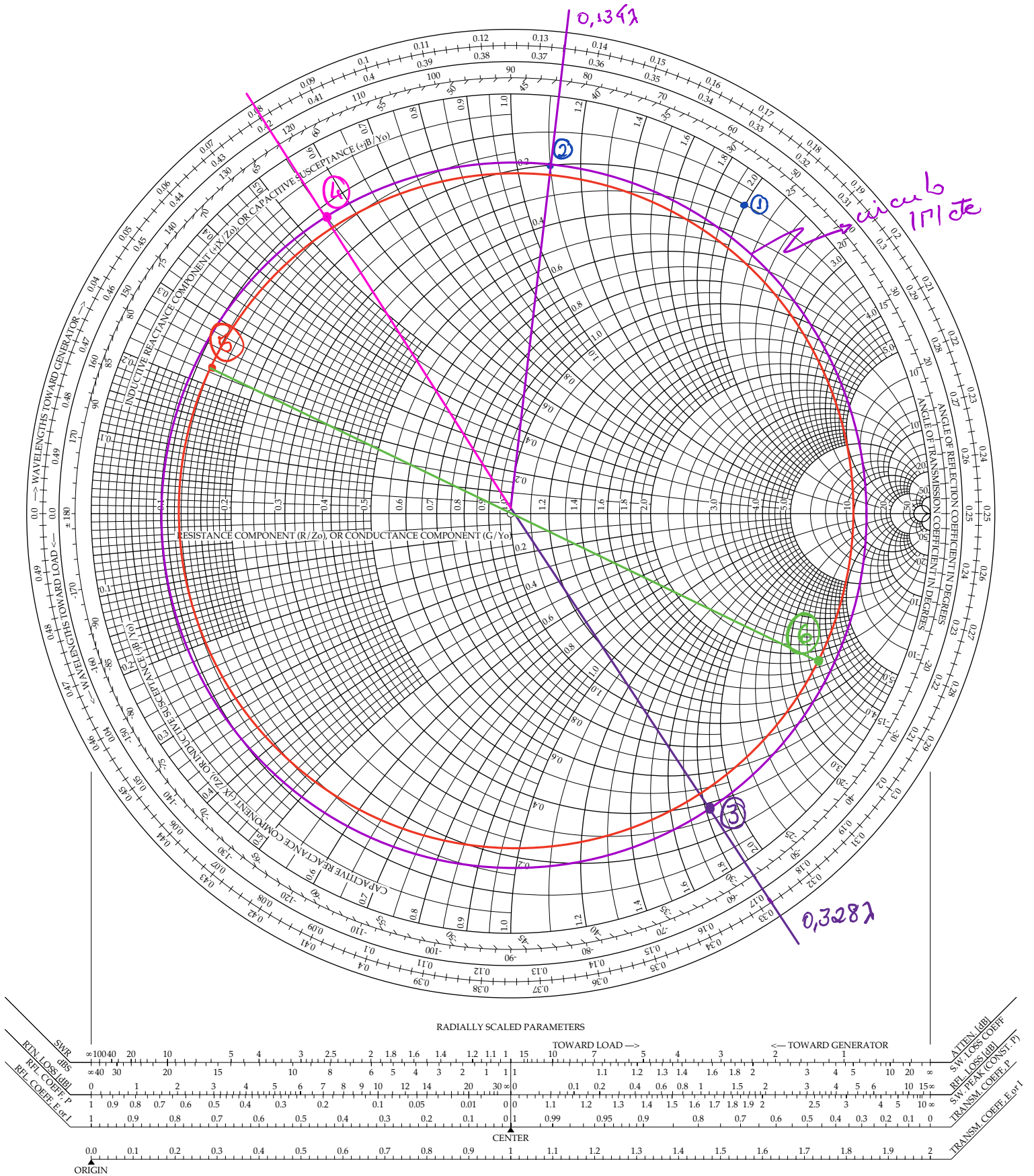


The Complete Smith Chart

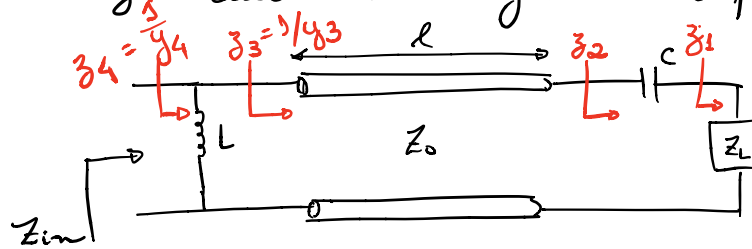
Black Magic Design



Questão Desafio :

Determine a impedância vista no circuito mostrado ao lado a $1,25 \text{ GHz}$ dadas as seguintes informações :

$$\begin{aligned} L &= 19,6 \text{ nH} \\ Z_0 &= 50 \Omega \\ l &= 0,194\lambda \\ C &= 2,83 \text{ pF} \\ Z_L &= 10 + 100j \end{aligned}$$



$$Z_L = 10 + 100j$$

$$y_L = \frac{Z_L}{Z_0} = \frac{10 + 100j}{50} = 0,2 + 2j$$

$$X_C = \frac{-j}{\omega C} = \frac{-j}{2\pi \times 1,25 \times 10^9 \times 2,83 \times 10^{-12}} = -j 45$$

$$x_c = \frac{-j \cdot 45}{50} = -j \cdot 0,9$$

$$0,2 + 2j - 0,9j = 0,2 + 1,1j$$

comprimento em ② + l = comprimento em ③

$$0,134\lambda + 0,194\lambda = 0,328\lambda$$

$$y_L = \frac{-j Z_0}{\omega L} = \frac{-j \cdot 50}{2\pi \cdot 1,25 \cdot 10^9 \cdot 19,6 \cdot 10^{-9}} = -j \cdot 50 \cdot 0,0065 = -j 0,325$$

$$y_{in} = y_3 + y_L = 0,12 + 0,55j - 0,325j = 0,12 + 0,225j$$

$$z_4 = z_{in} = 1,8 - 3,6j$$

$$Z_{in} = (1,8 - 3,6j) \cdot 50 = 90 - 180j$$

$$Z_{in} = 90 - 180j$$

① $0,2 + 2j$ (z_1)

② $0,2 + 1,1j$ (z_2)

③ $0,4 - 1,8j$ (z_3)

④ $0,12 + 0,55j$ (y_3)

⑤ $0,12 + 0,225j$ (y_4)

⑥ $1,8 - 3,6j$ (z_4)